

End-semester Exam Solutions

Q1. $D=1$, if $P=1$, $E=0$ & $K=1$ or $M=1$

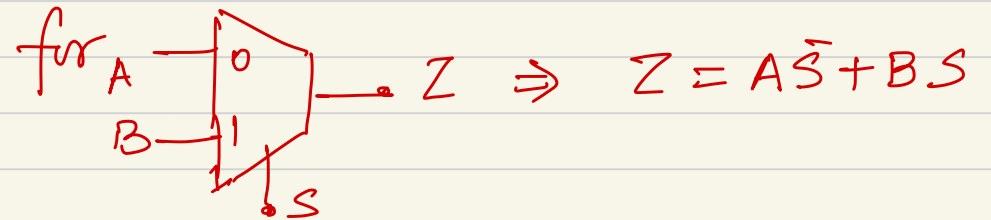
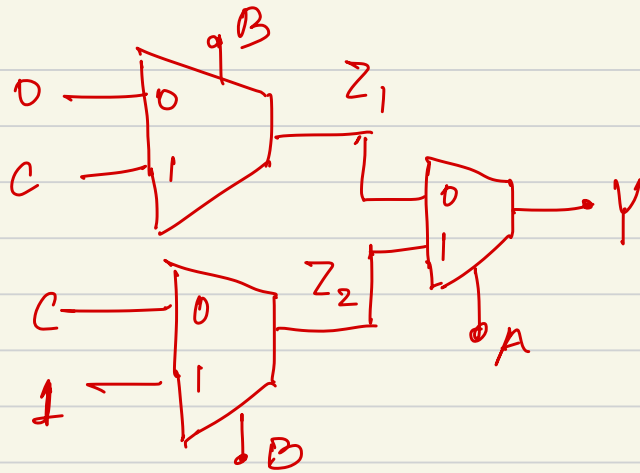
$$D = P\bar{E}(K+M)$$

Q2. A B C Z
 $m_1, m_3, m_5,$
 don't $\rightarrow m_6, m_7$
 care

AB		00	01	11	10
C	0			X	
	1	1	1	X	1

$\Rightarrow Z = C$

Q3.



$$\Rightarrow Z_1 = 0 \cdot \bar{B} + BC = BC$$

$$Z_2 = \bar{B} \cdot C + B \cdot 1 = B + \bar{B}C = B + C$$

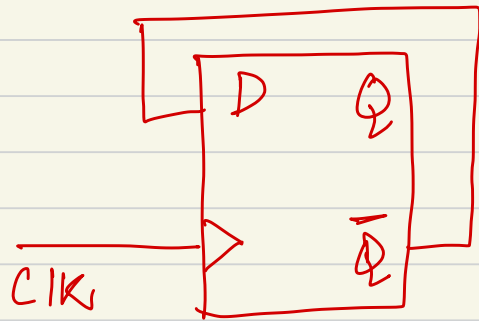
$$\Rightarrow Y = Z_1 \bar{A} + Z_2 A = BC \bar{A} + AB + AC$$

$$= C(A + \bar{A}B) + AB = C(A + B) + AB$$

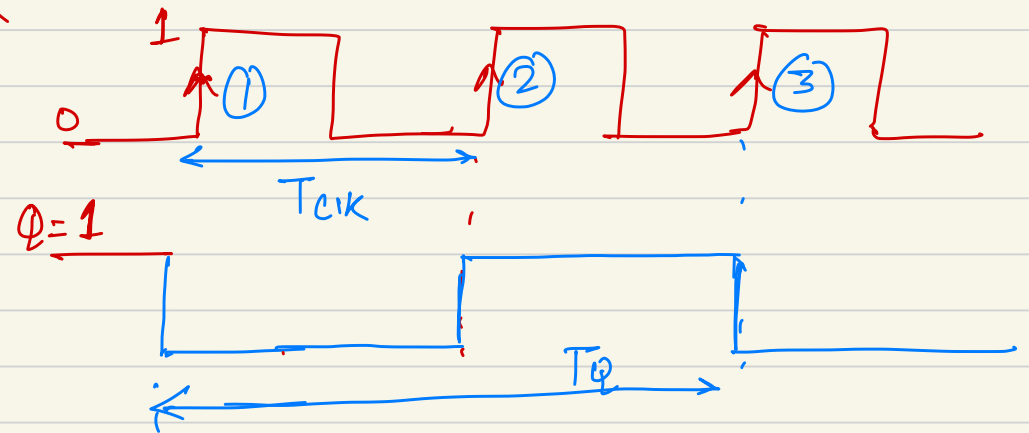
$$\boxed{Y = AC + BC + AB}$$

For Y to be 1, at least $A \& B$ or $B \& C$ or $A \& C$ needs to be 1.
i.e. minimum two inputs need to be 1.

Q4.



CLK



at first clock edge 1

$$D=0, \Rightarrow Q=0 \Rightarrow \bar{Q}=1$$

at ②

$$D=1, \Rightarrow Q=1 \Rightarrow \bar{Q}=0$$

at ③

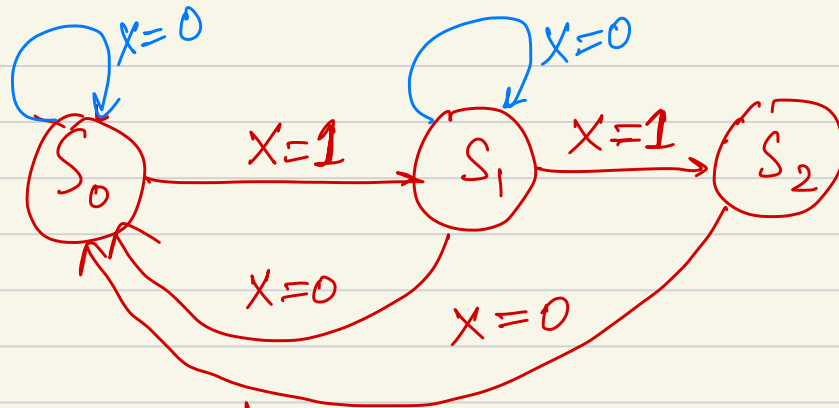
$$D=0 \Rightarrow Q=0 \Rightarrow \bar{Q}=1$$

if $\bar{Q}=1$
 $\Rightarrow Q=0$
 $\Rightarrow D=0$

$$\text{frequency of } Q = \frac{f_{CLK}}{2}$$

$$\text{as } T_Q = 2 T_{CLK}$$

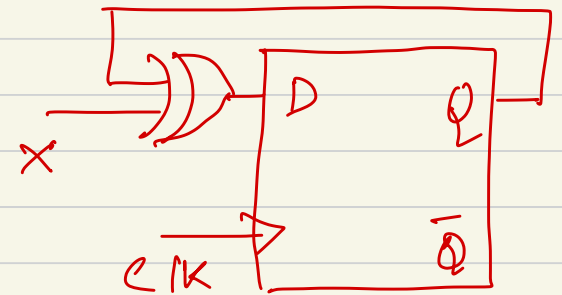
Q5.



As there are only three states, they can be represented with two bit state output i.e 2 D Flip Flops.

Q6.

X	$Q(t)$	D	$Q(t+1) = D$
0	0	0	0
0	1	1	1
{ 1	0	1	1
	1	0	0



When $X = 1$, at each clock edge the next state flips or toggles.

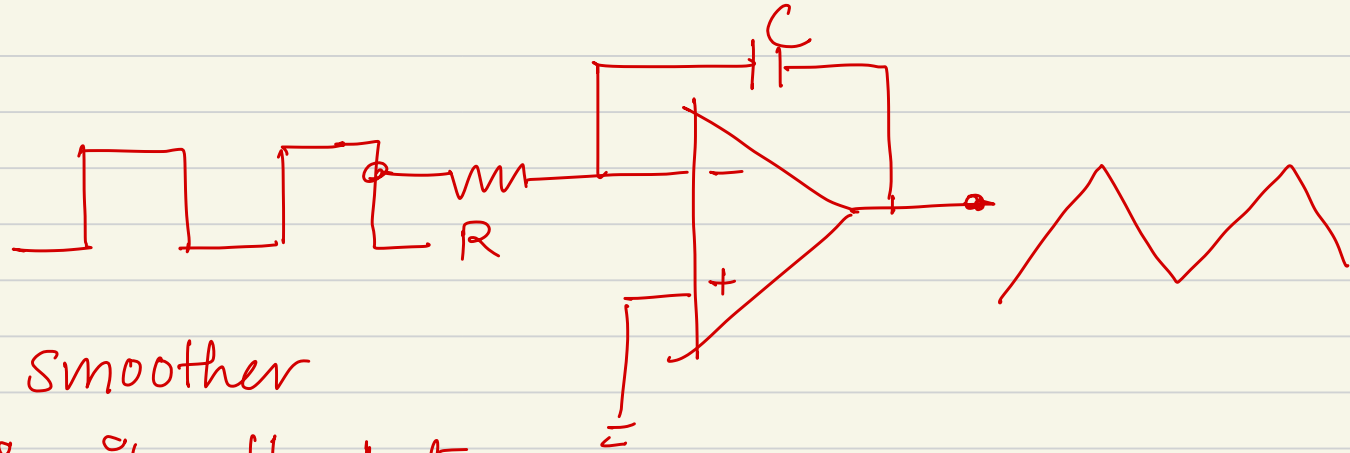
Q7. resolution = $0.5V$, reference voltage = $5V$,
levels = $\frac{5}{0.5} = 10$, i.e. minimum no. of bits
 $n > \log_2(10) \Rightarrow \underline{n = 4}$

Q8. if MSB resistance is R , weight of MSB = 2^{N-1} , weight of LSB = 1
as the weight of LSB is lower, $R_{LSB} \gg R_{MSB}$
 $\Rightarrow \underline{R_{LSB} = R \cdot 2^{N-1}}$

Q9. For $10\ \Omega$ motor, if $5V$ is desired $\Rightarrow$ current supplied to
load = $\frac{5}{10} = 0.5A$ or $500mA$.

As OPamp can only give a maximum of $20mA$, we don't achieve $5V$ output

Q10:



the output becomes smoother
as the integrator circuit attenuates
the high frequency components in the square wave.

Thus, triangular wave represents the information lost (ie. high frequency components) specially contributing to absence of sharp transitions.

Q11. A B C $\leftarrow$ sensors

$Y = 1$, when $A \& B = 1$ or $B \& C = 1$ or $A \& C = 1$

A) Truth Table

	A	B	C	Y
m_0	0	0	0	0
m_1	0	0	1	0
m_2	0	1	0	0
m_3	0	1	1	1
m_4	1	0	0	0
m_5	1	0	1	1
m_6	1	1	0	1
m_7	1	1	1	1

B)

Y

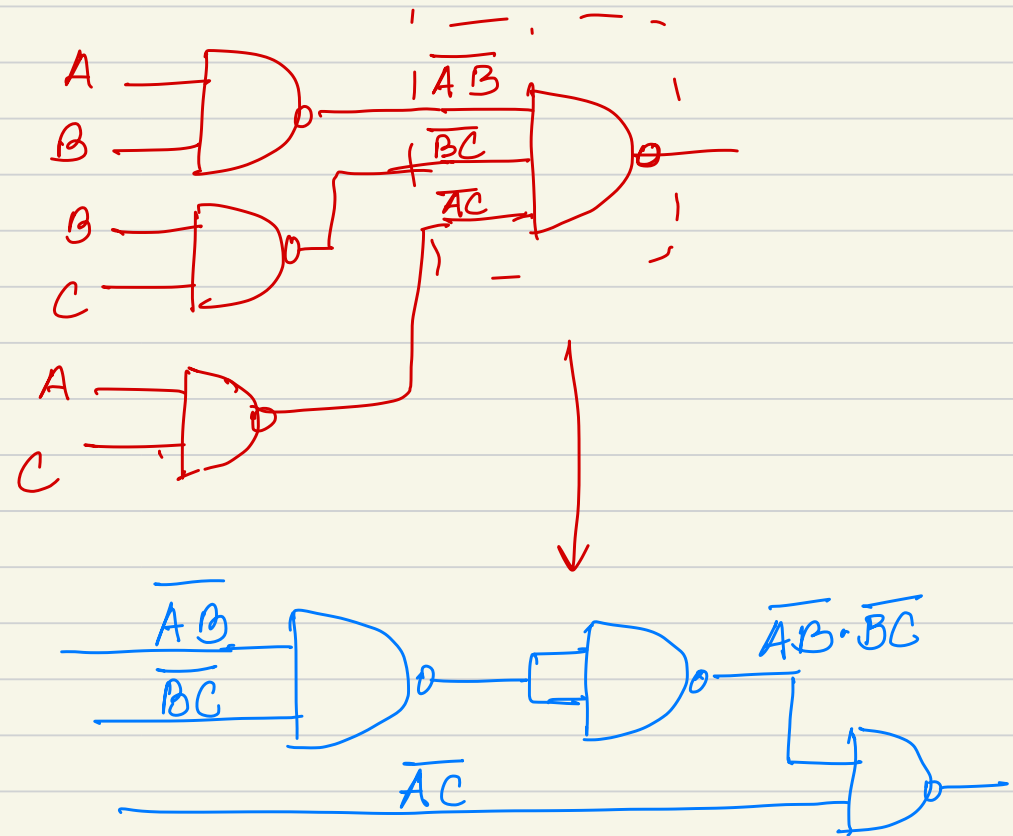
	AB			
	00	01	11	10
C				
0			1	
1		1	1	1

$$\Rightarrow Y = AB + BC + AC$$

C) $Y = AB + BC + AC$

using NAND gates (2-input)

$$Y = \overline{\overline{AB + BC + AC}} = \overline{\overline{AB} \cdot \overline{BC} \cdot \overline{AC}}$$



A)

Q_2	Q_1	Q_0	D_2	D_1	D_0
0	0	0	0	0	1
0	0	1	0	1	1
0	1	1	0	1	0
0	1	0	1	1	0
1	1	0	1	1	1
1	1	1	1	0	1
1	0	1	1	0	0
1	0	0	0	0	0

B)

Q_2	Q_1	Q_0	D_2
0	0	0	0
0	0	1	1
0	1	1	1
0	1	0	0
1	1	0	1
1	1	1	1
1	0	1	0
1	0	0	0

$$D_2 = Q_1 \bar{Q}_0 + Q_2 Q_0$$

Q_2	Q_1	Q_0	D_1
0	0	0	0
0	0	1	1
0	1	1	1
0	1	0	0
1	1	0	1
1	1	1	1
1	0	1	0
1	0	0	0

$$D_1 = \bar{Q}_2 \bar{Q}_0 + Q_1 \bar{Q}_0$$

D_0

Q_2	Q_1	Q_0	D_0
0	0	0	1
0	0	1	1
0	1	1	0
0	1	0	0
1	1	0	1
1	1	1	1
1	0	1	0
1	0	0	0

$$D_0 = \bar{Q}_2 \bar{Q}_1 + Q_2 Q_1 = \overline{Q_2 \oplus Q_1}$$

C)

For the given circuit Fig 3;

$$D_0 = \overline{Q_1 \oplus Q_2}, \quad D_1 = Q_1 Q_0 + \overline{Q_2} Q_0$$

$$D_2 = Q_2 Q_0 + Q_1 \overline{Q_0}$$

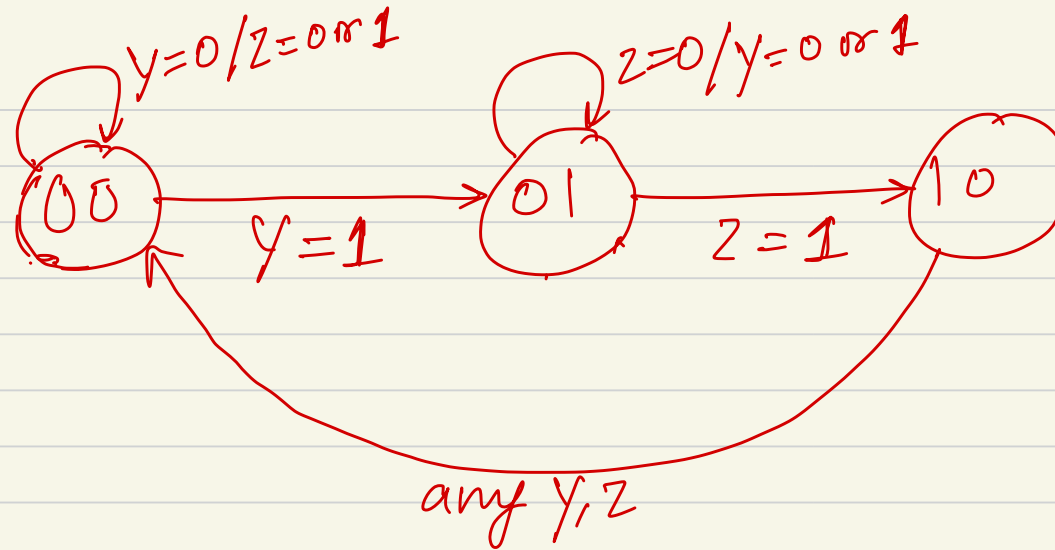
Q_2	Q_1	Q_0	D_2	D_1	D_0
0	0	0	0	0	1
0	0	1	0	1	1
0	1	1	0	1	0
0	1	0	$\nrightarrow$ 1	0	0
1	0	0	0	0	0

No, it does not give the intended sequence.

* Comparing with D equations derived in Part B:-

' D_1 ' equation is wrong where $Q_1 Q_0$ should be $Q_1 \overline{Q_0}$

Q.13.



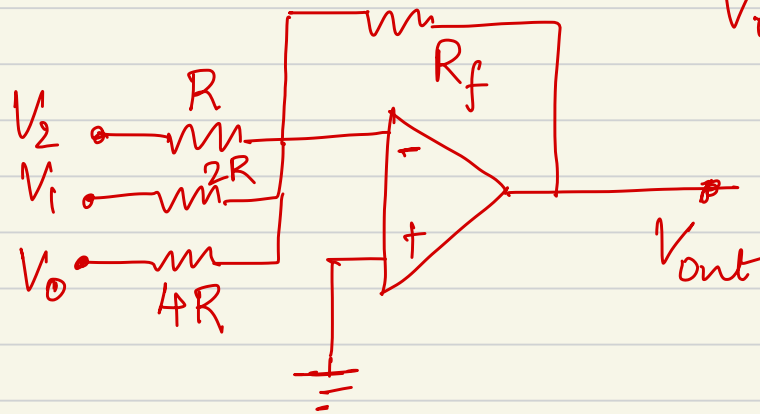
Q_1 Q_0
for representing
states

Q_1	Q_0	Y	Z	D_1	D_0
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	1	0
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	1	1	0

Q_1	Q_0	Y	Z	D_1	D_0
1	0	0	0	0	0
1	0	0	1	0	0
1	0	1	0	0	0
1	0	1	1	0	0
1	1	0	0	X	X
1	1	0	1	X	X
1	1	1	0	X	X
1	1	1	1	X	X

Q14

3-bit binary weighted DAC



taking $V_2 = Q_2 V_{ref}$, $V_1 = Q_1 V_{ref}$ & $V_0 = Q_0 V_{ref}$
 $V_{ref} = 5V$

$$V_{out} = -V_{ref} \left(\frac{R_f}{R} Q_2 + \frac{R_f}{2R} Q_1 + \frac{R_f}{4R} Q_0 \right)$$

$$\Rightarrow V_{out} = -\frac{V_{ref} \cdot R_f}{R} \left(Q_2 + \frac{Q_1}{2} + \frac{Q_0}{4} \right)$$

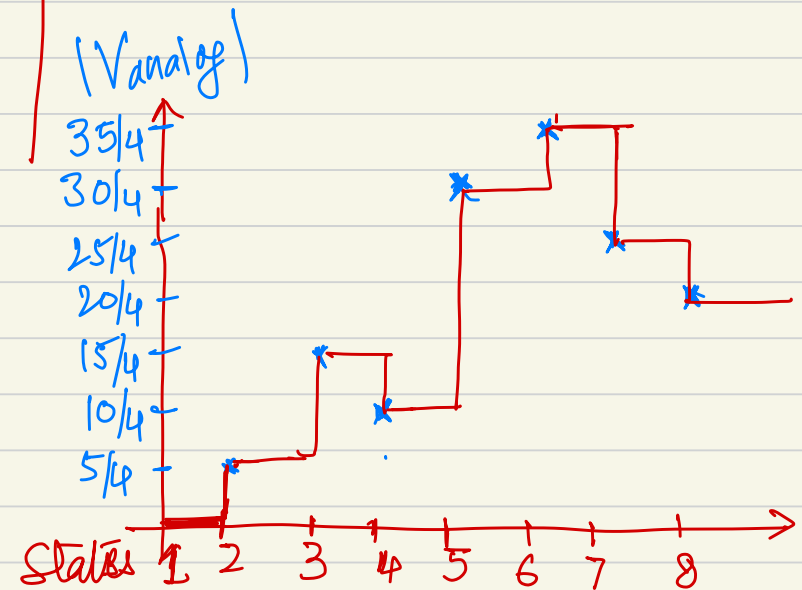
if $R_f = R$,

$$\Rightarrow V_{out} = -5V \cdot \left(Q_2 + \frac{Q_1}{2} + \frac{Q_0}{4} \right)$$

$$\text{for } 011 \Rightarrow V_{out} = -5V \cdot \left(\frac{1}{2} + \frac{1}{4} \right) = -\frac{3 \times 5}{4} V = -\frac{15}{4} V$$

for Q12 sequence

		Q_2, Q_1, Q_0	V_{analog}	$ V_{analog} $
State 1		0 0 0	0	0
" 2		0 0 1	$-5/4$	$5/4$
" 3		0 1 1	$-15/4$	$15/4$
" 4		0 1 0	$-10/4$	$10/4$
" 5		1 1 0	$-30/4$	$30/4$
" 6		1 1 1	$-35/4$	$35/4$
" 7		1 0 1	$-25/4$	$25/4$
" 8		1 0 0	$-20/4$	$20/4$



Q15.

$$\frac{V_{out}}{V_{in}} = - \frac{\frac{R_2 \times 1/sC_1}{R_2 + 1/sC_1}}{R_1} = - \frac{R_2}{R_1} \cdot \frac{1}{1 + sC_1 R_2}$$

A. Cutoff frequency, $f_c = \frac{1}{2\pi R_2 C_1} = \frac{1 \times 100}{2\pi} = 15.91 \text{ Hz}$

B. If the input frequency is much higher than f_c , high frequency components in step transitions will be attenuated.

Similar to Q10, the transitions will be smoothed.

